



Module Title

Cybersecurity and Networking

Internship Assignment Submission

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1. Questions And Answers

1. What is a Network Interface Card (NIC)?

- A Network Interface Card (NIC) is a hardware component that allows a computer or device to connect to a network. It can provide connectivity through Ethernet cables or Wi-Fi.

2. What is Ethernet?

- Ethernet is a widely used technology for connecting devices in a wired local area network (LAN). It allows devices such as computers, switches, and routers to communicate with each other.

3. What is RJ-45?

- RJ-45 is a common connector used for connecting Ethernet network cables to devices such as computers, switches, and routers. It usually has eight pins.

4. What does UTP stand for?

- UTP stands for Unshielded Twisted Pair. It is a type of copper networking cable containing pairs of twisted wires and is commonly used in Ethernet networks.

5. Name three common Ethernet cable categories.

- Three common Ethernet cable categories are:
 - Cat5e
 - Cat6
 - Cat6a

6. What is a straight-through cable?

- A straight-through cable has the same wiring arrangement at both ends. Traditionally, it was mainly used to connect different types of devices, such as a PC to a switch.

7. What is a crossover cable?

- A crossover cable has different wiring arrangements at each end, allowing the transmit and receive connections to cross over. Traditionally, it was used to directly connect similar devices, such as PC to PC or switch to switch.

8. Explain Auto MDI-X.

- Auto MDI-X is a feature that automatically detects how the network cable is connected and adjusts the transmit and receive signals correctly. Because of this technology, modern devices usually work with either straight-through or crossover cables.

9. What is the difference between copper and fiber-optic networking?

Copper Networking	Fiber-Optic Networking
Uses electrical signals	Uses light signals
Usually cheaper	Usually more expensive
Supports shorter distances	Supports much longer distances
Can be affected by EMI	Not affected by electromagnetic interference
Can support PoE (Power over Ethernet)	Does not normally support PoE in the same way

10. What is multimode fiber?

- Multimode fiber is a type of fiber-optic cable designed mainly for shorter-distance communication. It is commonly used in places such as buildings, campuses, and data centers.

11. What is single-mode fiber?

- Single-mode fiber is designed for transmitting data over long distances. It is commonly used by telecommunication companies, ISPs, and large network infrastructures.

12. Give two advantages of fiber-optic networking.

- Two major advantages are:
 1. It can transmit data over very long distances.
 2. It provides high bandwidth and is not affected by electromagnetic interference (EMI).

13. Why is understanding network interfaces important for cybersecurity?

- Understanding network interfaces is important because cybersecurity professionals need to know how devices connect and communicate on a network. This helps them monitor network traffic, identify unauthorized devices, configure security controls, troubleshoot problems, and detect possible attacks.

2. Practical Identification

14. Identify your active network adapter.

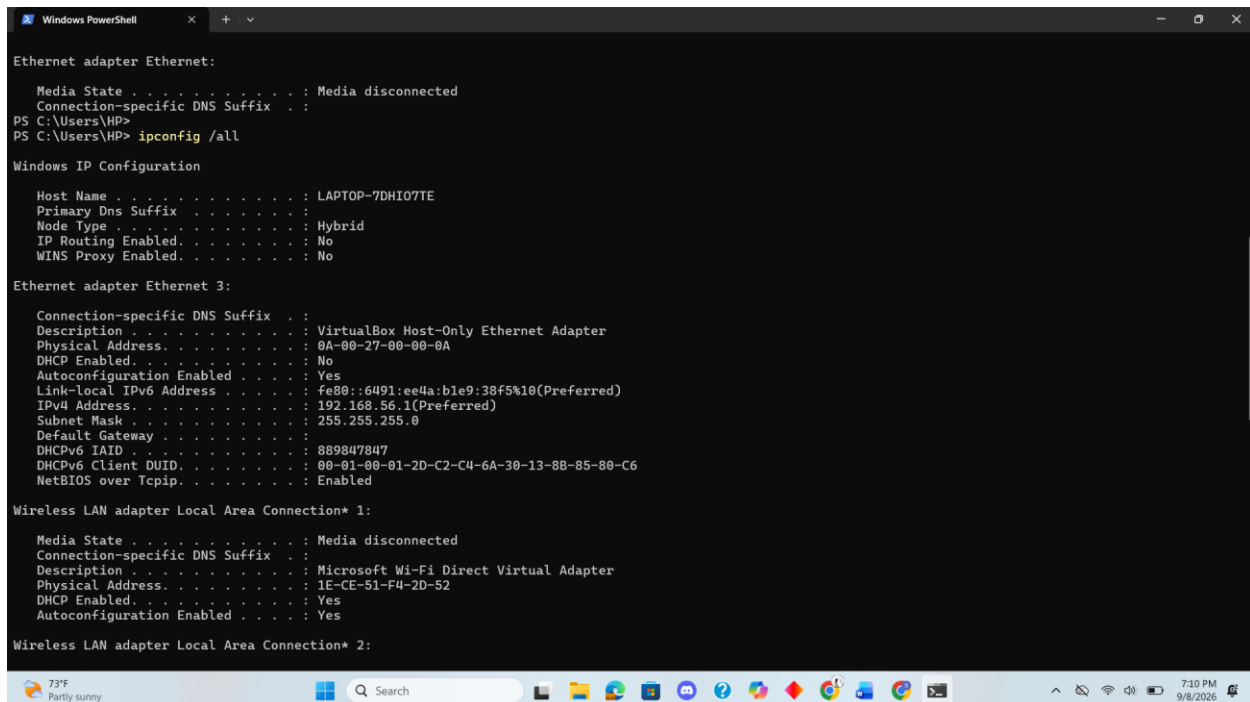
- My active network adapter is: Wi-Fi

It is active because it has an IPv4 address and a default gateway.

15. Write your IPv4 address, default gateway, and network adapter name.

- **Network Adapter Name:** Wi-Fi
- **IPv4 Address:** 192.168.101.12
- **Default Gateway:** 192.168.101.1

3. Screenshots



```
Windows PowerShell
Ethernet adapter Ethernet:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 
PS C:\Users\HP> ipconfig /all
Windows IP Configuration

Host Name . . . . . : LAPTOP-7DHI07TE
Primary Dns Suffix . . . . . : 
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 3:
    Connection-specific DNS Suffix . : 
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter
    Physical Address. . . . . : 0A-00-27-00-00-0A
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . : fe80::6491:ee4a:b1e9:38f5%10(Preferred)
    IPv4 Address. . . . . : 192.168.56.1(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 
    DHCPv6 IAID . . . . . : 889847847
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-00-C6
    NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
    Physical Address. . . . . : 1E-CE-51-F4-2D-52
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:
```

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. : 12-CE-51-F4-2D-52
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
Physical Address. : 1C-CE-51-F4-2D-52
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
IPv6 Address. : 2404:7c00:48:35a3:a2:27ca:6356:d6f7(Preferred)
Temporary IPv6 Address. : 2404:7c00:48:35a3:5dbe:3de7:588f:ab64(Preferred)
Link-local IPv6 Address : fe80::5305:27d1:a329:219d%5(Preferred)
IPv4 Address. : 192.168.101.12(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : Tuesday, September 8, 2026 11:09:36 AM
Lease Expires : Wednesday, September 9, 2026 5:24:42 PM
Default Gateway : fe80::72a5:6aff:feb3:259a%5
 192.168.101.1
DHCP Server : 192.168.101.1
DHCPv6 IAID : 85773905
DHCPv6 Client DUID. : 00-01-00-01-2D-C2-C4-6A-30-13-8B-85-80-C6
DNS Servers : fe80::1%5
 110.44.112.245
 110.44.113.245
 fe80::1%5
NetBIOS over Tcpip. : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : 1C-CE-51-F4-2D-53